

The boards we've reviewed are based on the most feature-rich of all the chipsets—the nForce4 Ultra and nForce4 SLI, since the other chipsets, even with the good performance and features they may be offering, are priced lower, and hence are confined to the budget or mid-range segment. The only extra feature the nForce4 SLI has over the nForce4 Ultra is support for two PCIe x16 graphics card in SLI mode.

Features

Like we said, all the motherboards we received were based on either nForce4 Ultra or nForce4 SLI. Hence, every board supported the AMD Athlon64 X2, which was what we used to test the motherboards with.

There is no variation in the type and speed of memory in this segment: the supported memory was DDR 400 MHz. All the motherboards supported SATA II, since the nForce4 chipset supports this standard. Similarly, most of the motherboards supported the SATA hot-plug function, which enables you to remove or plug in a SATA drive without powering off the computer!

The motherboards from DFI and Gigabyte, as well as the ASUS A8N-SLI, all had eight SATA ports, so that up to eight SATA drives can be added to the system. To complement the number of SATA ports, the Gigabyte GA-K8NXP-SLI and the ASUS A8N-SLI came bundled with eight SATA cables.

The MSI K8N SLI Platinum sports an on-board Creative SoundBlaster Live! 24-bit audio solution, which is the best in this class. The DFI LanParty UT NF4 SLI-D features the 'Karajan' module, which is a detachable audio module that isolates audio signals so as to produce the best quality audio. Some cabinets, however, might not be able to accommodate the board with this module.

Dual LAN—a feature likely to appeal to gamers—was available on most of the mother-



Gigabyte 8Σ Series GA-K8 NXP-SLI

boards. Computers on a LAN can connect to each other via one LAN port, while the other LAN ports are all connected to the Internet. A team of gamers on a LAN can thus play online with gamers around the world!

A Wi-Fi 802.11 b/g card was bundled with the Gigabyte 8Σ Series GA-K8NXP-SLI to facilitate wireless connectivity with access points at hotspots. A U-Plus DPS (Dual Power System) card was also bundled with this board. This card aids in stability of the system in the face of power fluctuations. It also has LEDs that indicate the power and load status.

The DFI LanParty motherboard had UV reactive slots and round cables. The cables and slots emit a lovely glow when UV light falls on them. This might not be visible if ordinary cabinets are used, but will be very visible if a modded cabinet with transparent panels is used. Of course, you're going to need to buy yourself a UV cold-cathode light for the glow to happen.

How We Tested

Our benchmarks and tests were chosen after careful consideration so as to stress the boards to the maximum. As many as five test-bench setups had to be used because of the various types of processor sockets available on the motherboards we tested.

Intel motherboard setups featured the Pentium IV 3.6 GHz processor, while AMD motherboard setups featured the Athlon 64 X2 dual-core processor. Other than that, the test setups consisted of Corsair 1 GB DDR 400 MHz RAM or Micron 1 GB DDR2 533 MHz RAM, according to the RAM supported by the motherboard; a 120 GB 7200 rpm SATA Maxtor hard drive; and an XFX NVIDIA GeForce 7800 GT PCI-Express display card. The systems were powered by Antec NEOPOWER 480 Watts power supply.

Windows XP Pro with SP1 was freshly loaded along with DirectX 9.0c, and the drivers from the driver CDs provided with the motherboards were loaded.

Our tests and benchmarks consisted of the following:

FarCry: This is a Direct3D game that accurately measures the Direct3D gaming subsystem. This game was chosen because it uses a fairly modern graphics engine, especially for rendering water, and represents how most games will play on your system in the future.

Doom 3: This OpenGL game is used to test the strength of the OpenGL graphics subsystem. We chose this game because it's very heavy on resources.

3DMark05: This is an industry-standard Direct3D synthetic benchmark from FutureMark, and it measures the graphics performance and capabilities of the system. This test gives you an indication about the graphics performance you can expect from the system.

PC Mark 2004: This is another industry-standard benchmark from FutureMark; it measures the CPU, memory, graphics and hard disk performance of the system—and hence is a good system-wide benchmark.

SiSoft Sandra 2005: This consists of benchmark modules that measure CPU, filesystem and memory performance.

Ziff-Davis Business Winstone 2004: This benchmark suite runs various applications such as MS Office 2002's Word, Excel, PowerPoint and Outlook, and other programs such as WinZip, to measure the real-world performance of the system.

Dr. DivX 1.06: This is a video encoding application used to encode DivX video files. A 100 MB DVD file was encoded to DivX, and we noted the time taken to encode.

We ran all the above benchmarks and logged all the scores. After applying the proper weightage to each of the logged scores, the aggregate performance score was calculated. Similarly, the various features of the motherboards, along with their prices, were noted to calculate the features and price scores. These, along with the performance scores, were taken into consideration to decide the winners.